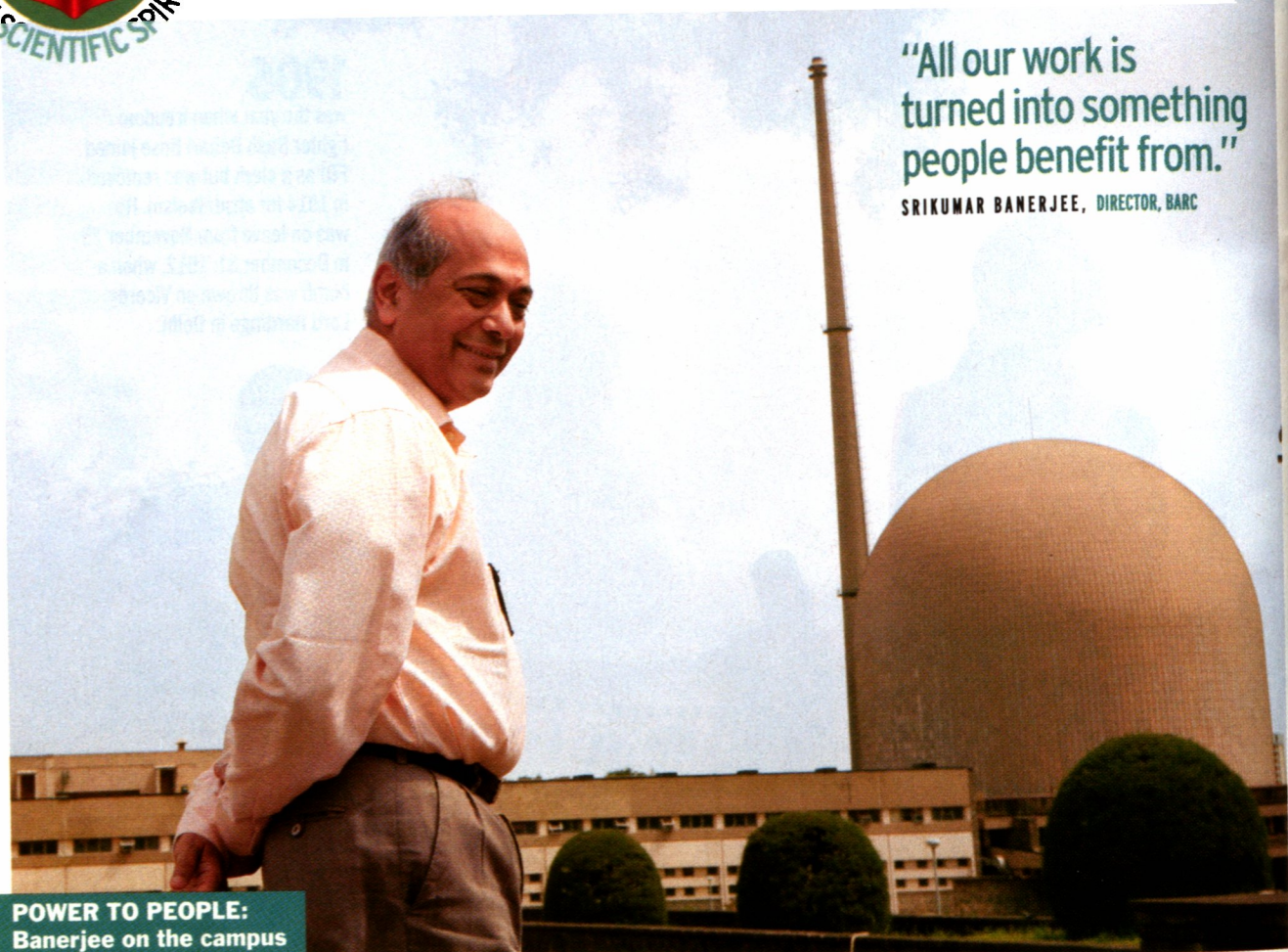




"All our work is turned into something people benefit from."

SRIKUMAR BANERJEE, DIRECTOR, BARC

POWER TO PEOPLE:
Banerjee on the campus

BHASKAR PAUL

BHABHA ATOMIC RESEARCH CENTRE, MUMBAI

NUCLEAR POWERHOUSE

From using gamma radiation for seeds to accelerate yield (1966) to developing a tsunami model to detect seismic activity (2005) and working on the Bhabhatron II machine for cancer treatment (2006), the Bhabha Atomic Research Centre (BARC) is at the forefront of the country's scientific development. "We don't sit in an ivory tower," says BARC Director Srikumar Banerjee, who has been with the organisation for the past four decades. The work can range from the incredible—immunising mangoes to make them export-friendly, turning kitchen waste into methane or creating an inspection gauge to detect faults in pipelines—to the expected, such as developing the Advanced Heavy Water Reactor (AHWR) which will enable power generation from thorium. Its series of firsts is inspiring—from developing the first supercom-

puter, Anupam, in 1991 to getting US certification for Krushak, the world's first cobalt-60 gamma radiation facility to be used in the treatment of mangoes. Its 6-km-long campus at Trombay houses reactors, training schools and a 1.5-km-long laboratory building. For those who work here, BARC is more than just a research institute. It is a centre which follows the three-D mantra: develop, design and deploy. "We are among the few countries to have mastered the entire nuclear fuel cycle starting from the mining of uranium ore to treating nuclear waste for safe disposal," says Banerjee. A top research project currently is high temperature reactors, which can split water molecules to get hydrogen to make synthetic fuel for automobiles. With 4,000 scientists and engineers, the campus is a powerhouse of ideas.

by Aditi Pa

NEVER GIVE IN

After the Pokhran experiment in 1974, when Canada pulled out, BARC scientists hung on and completed the Pressurised Heavy Water Reactor by themselves.